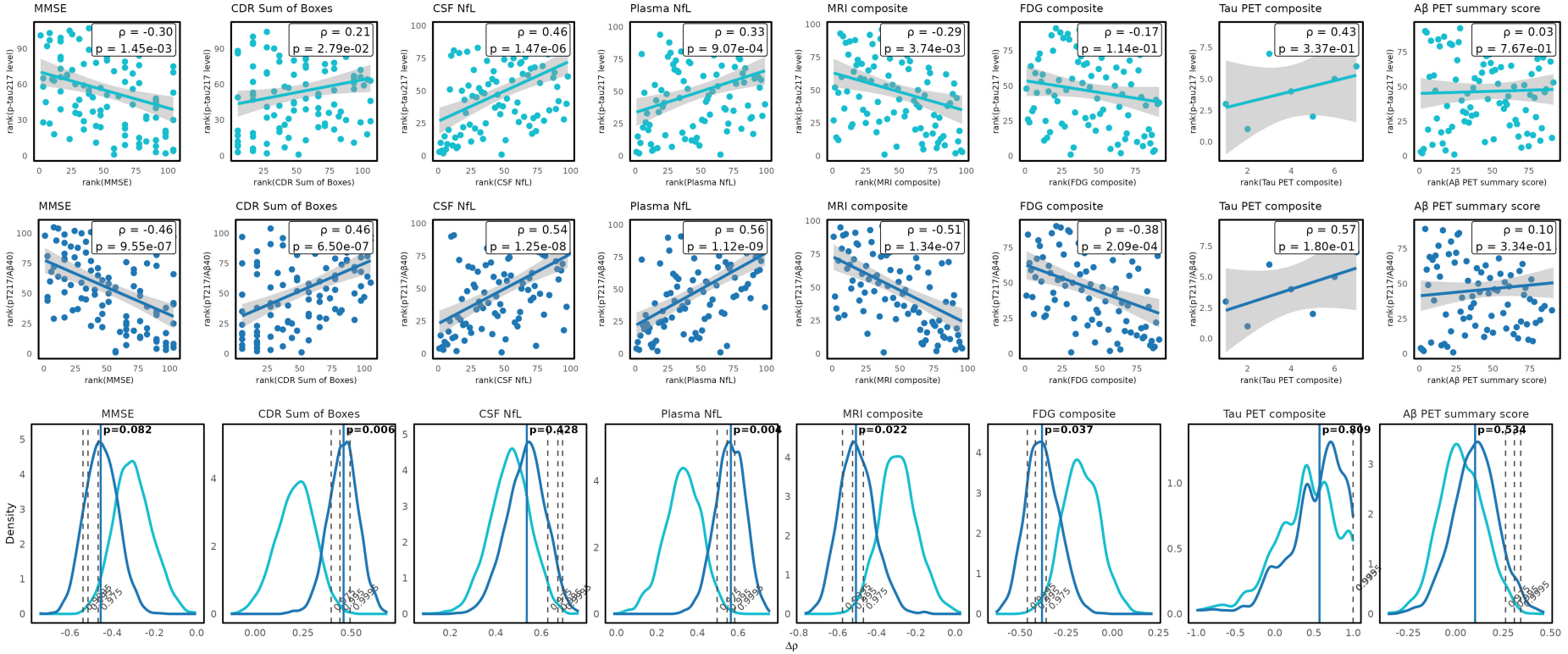
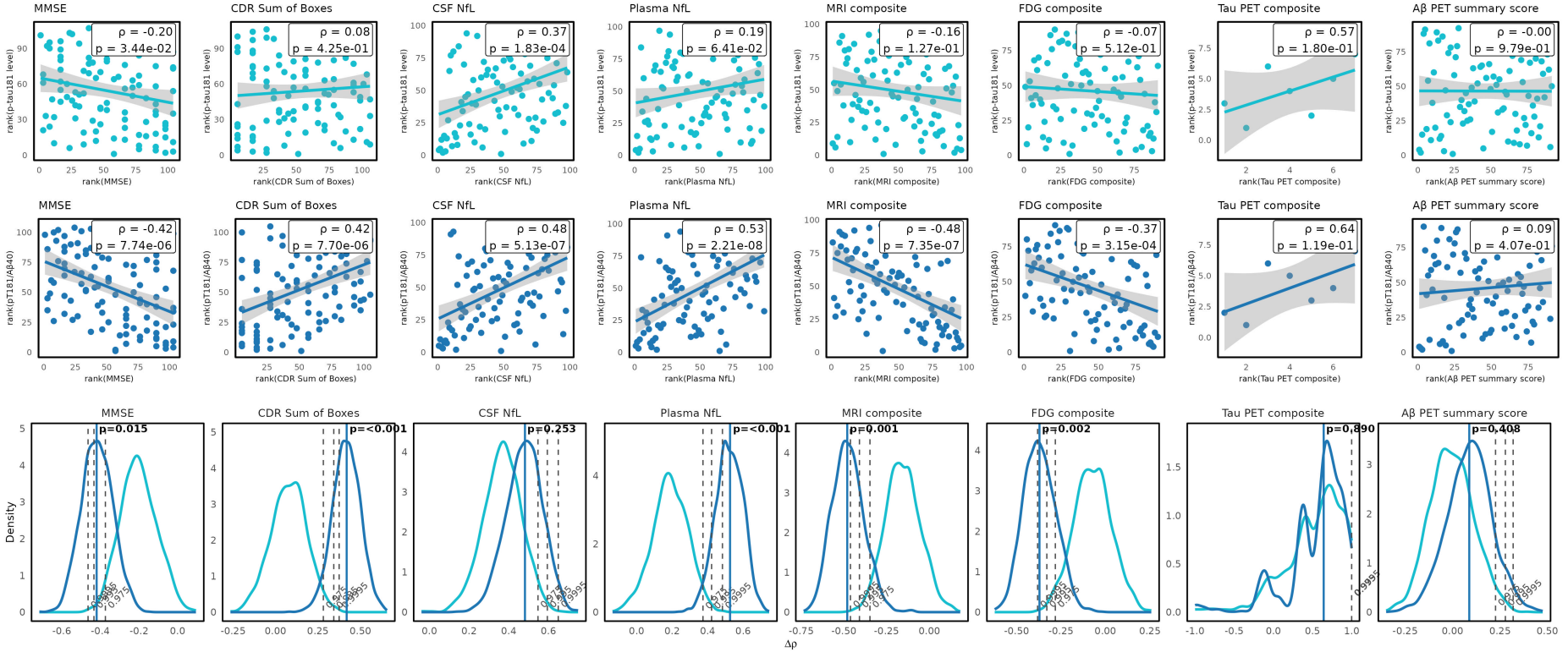
Supplementary material

Supplementary Fig. 3. Correlations of CSF biomarkers with clinical outcomes in Aβ+ sMC: raw vs. normalization to Aβ40.
Spearman correlations are shown between each fluid biomarker and cognitive, fluid, and imaging outcomes. Panels compare correlations computed using raw biomarker levels and Aβ40-normalized levels. The bottom panels display bootstrap distributions of Spearman's ρ for raw and Aβ40-normalized measures (2,000 resamples with replacement); dashed vertical lines mark the two-sided raw ρ quantiles (0.975/0.995/0.999) in the direction of the Aβ40-normalized shift, and the solid vertical line indicates the mean Aβ40-normalized ρ. The annotation reports the two-sided empirical p-value for the mean Aβ40-normalized ρ under the raw bootstrap distribution, before FDR correction.

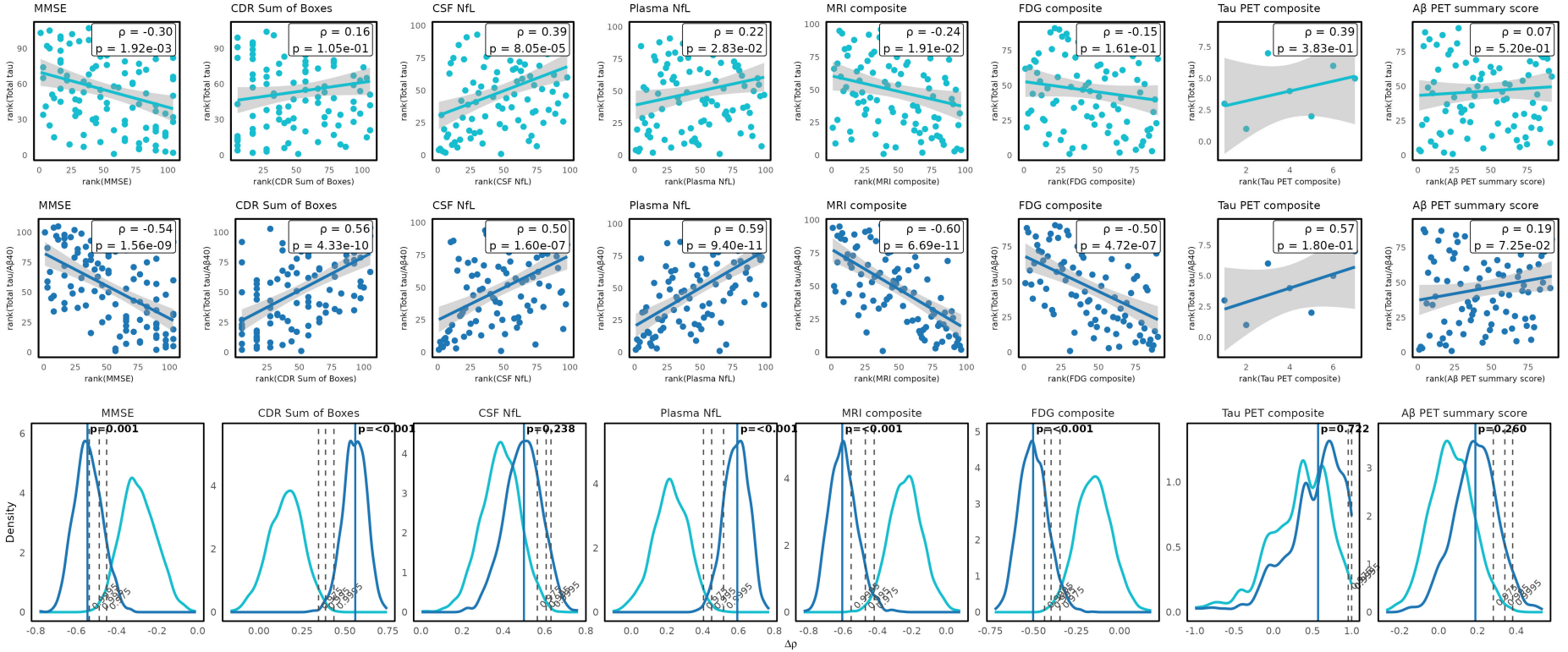
A) CSF p-tau217



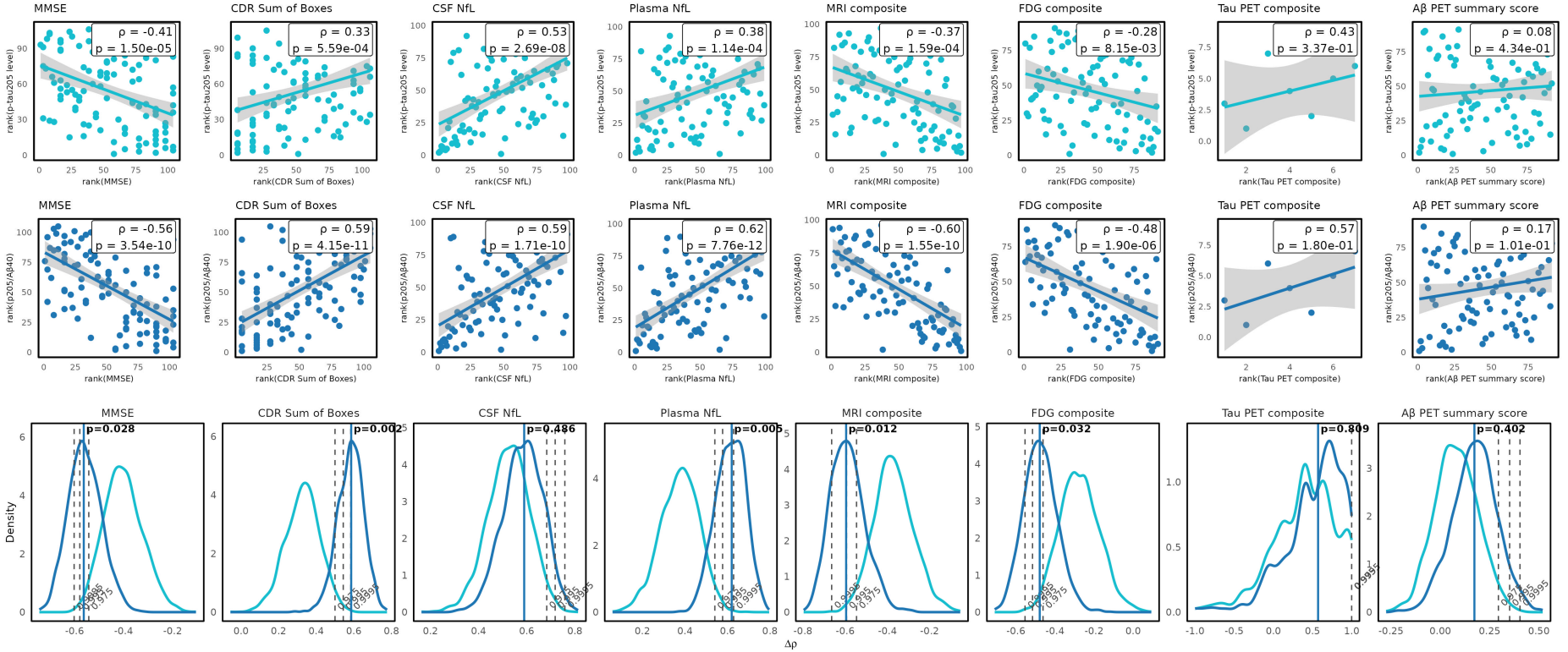
B) CSF p-tau181



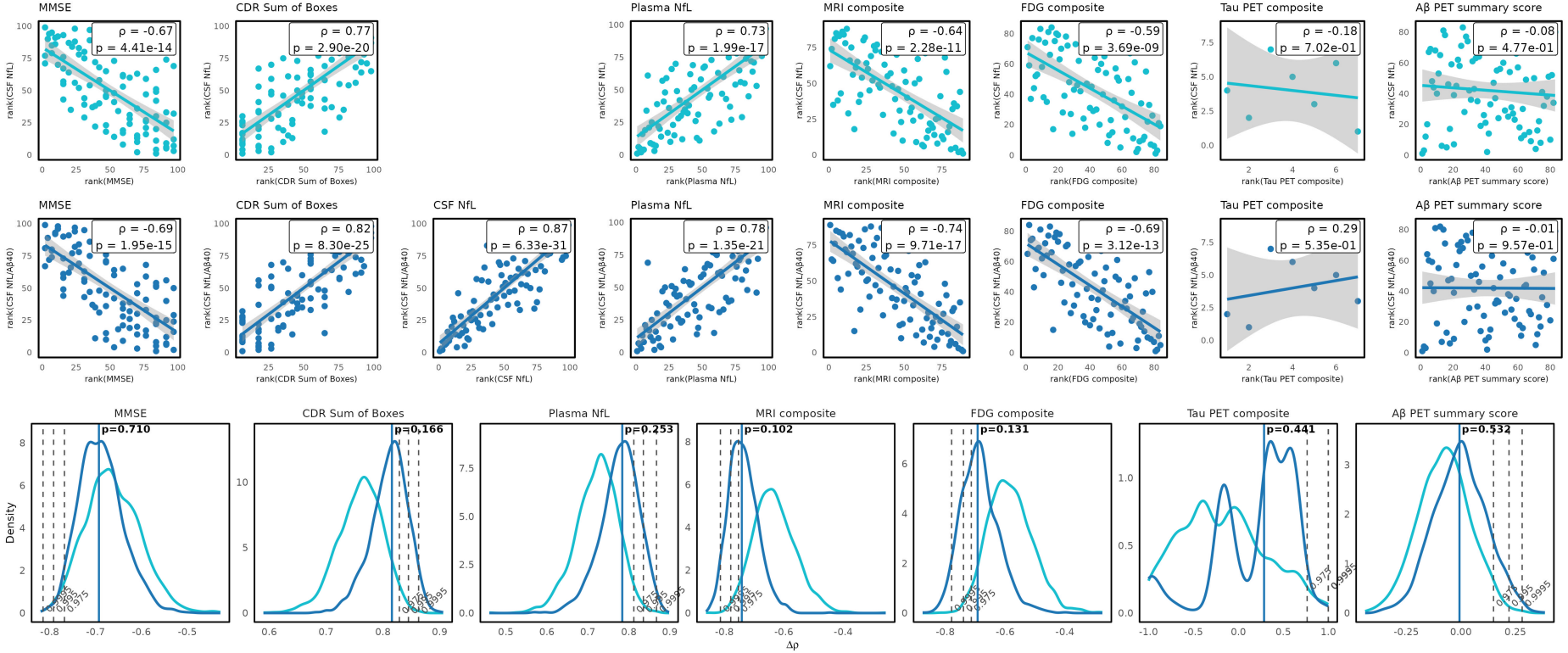
C) CSF Total tau



D) CSF p-tau205

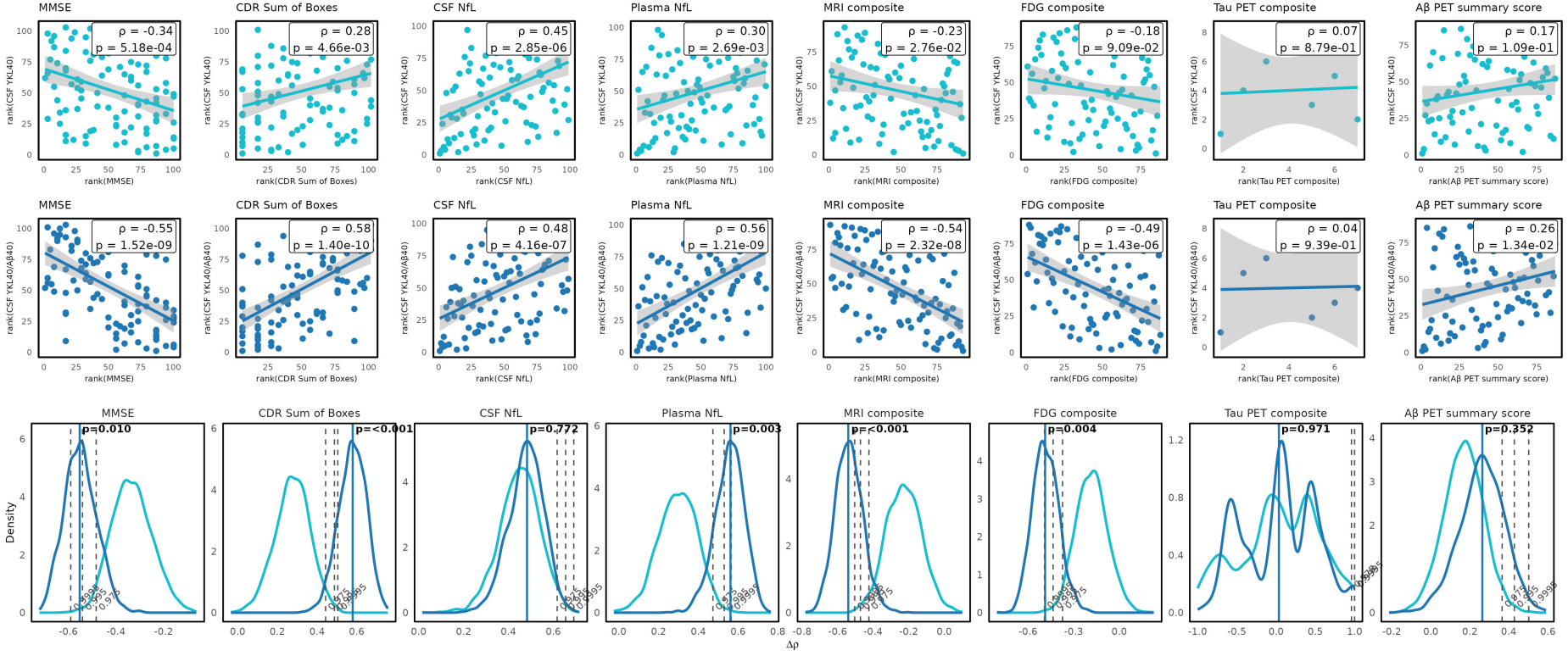


E) CSF NfL


MMSE
ρ = -0.67
p = 4.41e-14
rank(MMSE)
rank(CSF NfL)
CDR Sum of Boxes
ρ = 0.77
p = 2.90e-20
rank(CDR Sum of Boxes)
Plasma NfL
ρ = 0.73
p = 1.99e-17
rank(Plasma NfL)
MRI composite
ρ = -0.64
p = 2.28e-11
rank(MRI composite)
FDG composite
ρ = -0.59
p = 3.69e-09
rank(FDG composite)
Tau PET composite
ρ = -0.18
p = 7.02e-01
rank(Tau PET composite)
Aβ PET summary score
ρ = -0.08
p = 4.77e-01
rank(Aβ PET summary score)
MMSE
ρ = -0.69
p = 1.95e-15
rank(MMSE)
rank(CSF NfL/Aβ40)
CDR Sum of Boxes
ρ = 0.82
p = 8.30e-25
rank(CDR Sum of Boxes)
CSF NfL
ρ = 0.87
p = 6.33e-31
rank(CSF NfL)
Plasma NfL
ρ = 0.78
p = 1.35e-21
rank(Plasma NfL)
MRI composite
ρ = -0.74
p = 9.71e-17
rank(MRI composite)
FDG composite
ρ = -0.69
p = 3.12e-13
rank(FDG composite)
Tau PET composite
ρ = 0.29
p = 5.35e-01
rank(Tau PET composite)
Aβ PET summary score
ρ = -0.01
p = 9.57e-01
rank(Aβ PET summary score)

MMSE
p=0.710
CDR Sum of Boxes
p=0,166
Plasma NfL
p=0.253
MRI composite
p=0.102
FDG composite
p=0.131
Tau PET composite
p=0.441
Aβ PET summary score
p=0.532
Density
Δρ

F) CSF YKL-40


MMSE
ρ = -0.34
p = 5.18e-04
CDR Sum of Boxes
ρ = 0.28
p = 4.66e-03
CSF NfL
ρ = 0.45
p = 2.85e-06
Plasma NfL
ρ = 0.30
p = 2.69e-03
MRI composite
ρ = -0.23
p = 2.76e-02
FDG composite
ρ = -0.18
p = 9.09e-02
Tau PET composite
ρ = 0.07
p = 8.79e-01
Aβ PET summary score
ρ = 0.17
p = 1.09e-01
rank(CSF YKL40)
rank(MMSE)
rank(CDR Sum of Boxes)
rank(CSF NfL)
rank(Plasma NfL)
rank(MRI composite)
rank(FDG composite)
rank(Tau PET composite)
rank(Aβ PET summary score)
MMSE
ρ = -0.55
p = 1.52e-09
CDR Sum of Boxes
ρ = 0.58
p = 1.40e-10
CSF NfL
ρ = 0.48
p = 4.16e-07
Plasma NfL
ρ = 0.56
p = 1.21e-09
MRI composite
ρ = -0.54
p = 2.32e-08
FDG composite
ρ = -0.49
p = 1.43e-06
Tau PET composite
ρ = 0.04
p = 9.39e-01
Aβ PET summary score
ρ = 0.26
p = 1.34e-02
rank(CSF YKL40/Aβ40)
rank(MMSE)
rank(CDR Sum of Boxes)
rank(CSF NfL)
rank(Plasma NfL)
rank(MRI composite)
rank(FDG composite)
rank(Tau PET composite)
rank(Aβ PET summary score)
MMSE
p=0.010
CDR Sum of Boxes
p=<0.001
CSF NfL
p=0.772
Plasma NfL
p=0.003
MRI composite
p=<0.001
FDG composite
p=0.004
Tau PET composite
p=0.971
Aβ PET summary score
p=0.352
Density
Δρ